AN INTERNSHIP REPORT ON

“DATA VISUALISATION AND CIVIC SPATIAL ANALYTICS”

Carried out under the DS5P3 Data Visualisation Track (Team 1)

Submitted in Partial Fulfilment of the Requirements for the Award of the Degree of

BACHELOR OF COMPUTER APPLICATIONS (BCA — DATA SCIENCE)

of **Bengaluru North University**

Submitted By

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Bengaluru, Karnataka

Academic Year: 2026 – 2027

CERTIFICATE ISSUED BY THE ORGANIZATION / MENTOR

CERTIFICATE OF INTERNSHIP

This is to certify that **PITHAMBAR** bearing Register Number / USN: **U19XFS240052**, a student of **MVJ Degree College, Bengaluru**, pursuing **Bachelor of Computer Applications (BCA — Data Science)**, has successfully completed an internship program under the **DS5P3 Data Visualisation Track** from **17/10/2026 to 17/11/2026** (logging **92 hours** across 27 active days). During his internship, **PITHAMBAR** worked in **Team 1 — Data Visualisation & Civic Analytics** and gained hands-on experience in the following areas:

- ➢ Visual Perception Critique, Chart Selection, and Excel Dashboard Makeovers
- ➢ Data Cleaning, Merging, and Quality Auditing (Bengaluru Rainfall & Namma Metro Ridership)
- ➢ Power BI Star Schema Relationship Modelling, DAX Measures, and Report Page Design
- ➢ Python Data Visualisation (Matplotlib, Seaborn, Plotly), GeoJSON Choropleth Mapping, and Flask Backend Integration

His conduct, technical contribution, and collaborative discipline during his tenure with us were satisfactory. We wish him all the best for his future academic and professional endeavours.

Ajay Balamurugadas
Industry Mentor — DS5P3 Data Visualisation Track
[Authorized Seal & Signature]

Date: 17/11/2026
Place: Bengaluru

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CERTIFICATE

This is to certify that **PITHAMBAR** bearing Registered No. / USN: **U19XFS240052** is a bonafide student of **Bachelor of Computer Applications (BCA — Data Science)** of our College.

He has prepared the Internship Report entitled **"Data Visualisation and Civic Spatial Analytics"** (DS5P3 Data Visualisation Track, mentored by **Mr. Ajay Balamurugadas**) for the internship period from **17/10/2026 to 17/11/2026** towards the partial fulfilment of the requirements for the award of the **Bachelor of Computer Applications Degree of Bengaluru North University**.

Faculty Guide / Coordinator	**Head of Department**	**Principal**
Department of Computer Science	Department of Computer	MVJ Degree College
MVJ Degree College	Science	*[Seal & Signature]*

STUDENT DECLARATION

I, **PITHAMBAR**, Register Number / USN: **U19XFS240052**, hereby declare that this internship report entitled **“Data Visualisation and Civic Spatial Analytics”** is an original record of authentic work carried out by me during the internship period from **17/10/2026 to 17/11/2026** (92 hours) under the DS5P3 Data Visualisation Track, mentored by **Mr. Ajay Balamurugadas** and supervised by the Department of Computer Science, **MVJ Degree College, Bengaluru**.

I further declare that the datasets, workbooks, Python scripts, Power BI data models, and analytical visualisations documented in this report represent genuine team and individual deliverables submitted to the project repository, and that this report has not been previously submitted to any other university or institution for the award of any degree or diploma.

Signature: _____

Date: 17/11/2026 **Name: PITHAMBAR**

Place: Bengaluru **USN: U19XFS240052**

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PITHAMBAR (U19XFS240052)

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OVERVIEW

EXECUTIVE SUMMARY

This report documents the four-week intensive internship in **Data Visualisation and Civic Spatial Analytics** completed by **Pithambar (USN: U19XFS240052)** of **MVJ Degree College, Bengaluru**, as part of the **Bachelor of Computer Applications (BCA — Data Science)** final-year curriculum (DS5P3 Track) from **17 October 2026 to 17 November 2026**. Under the mentorship of **Mr. Ajay Balamurugadas**, Pithambar logged **92 hours across 27 active working days** collaborating within a three-member engineering team (Team 1: Pithambar, Nithin Kumar, and Vimal) using GitHub, VS Code, Microsoft Excel, SQL, Power BI, Python, Tableau, and Flask.

The internship followed a deliberate four-phase progression designed to build end-to-end competency from visual perception fundamentals to production-grade interactive web analytics:

- **Phase 1 — Visual Perception & Spreadsheet ETL (Days 1–8, 32 Hours):** Evaluated misleading 3D and cluttered charts versus clear time-series designs, authored a structured Question-to-Chart choice matrix, redesigned a multi-slice pie chart into an accurate horizontal bar chart (*pithambar_chart_make_over.xlsx*), and cleaned a 125-row historical Bengaluru rainfall series (1902–2025) incorporating El Niño and La Niña climate indicators alongside SQL fundamentals.
- **Phase 2 — Enterprise BI Modelling & Namma Metro Analytics (Days 9–13, 16 Hours):** Cleaned and merged multi-month BMRCL Namma Metro ridership workbooks (6,336 combined hourly rows, 26,211 August origin-destination rows with 1,305 aggregate rows isolated, and 166 station lookup mappings). Constructed Star Schema relationships in Power BI, authored three core DAX measures (*Average Ridership* = 365.073, *Peak Hour Ridership* = 5,376,931, and *Peak Hour Share* = 9.69%), and built Page 1 of the team dashboard covering 55,505,708 total rides across 84 stations.
- **Phase 3 — Python, Interactive Plotly & Civic GIS Mapping (Days 14–20, 22 Hours):** Documented Plotly Express and Streamlit workflows, built a Seaborn violin plot for Bengaluru house prices, exported an interactive Plotly correlation heatmap HTML, created an interactive BBMP ward-wise public goods choropleth using NFKD ward-name normalization, explored Tableau, and generated four interactive Plotly charts for the 198-ward Census 2011 Bengaluru Ward Story (8,443,675 residents).
- **Phase 4 — Capstone Platform: Lakes of Bengaluru (Days 21–27, 22 Hours):** Co-authored the one-page capstone charter, improved the Flask REST backend and KML/CSV data pipeline uniting 207 urban water bodies (5,390.4 acres across 8 civic zones; 51.7% rejuvenated, average WQI 64.3), and produced six publication-grade Matplotlib/Seaborn visuals proving strong inverse correlation (r = -0.84) between perimeter encroachment and biodiversity.

Key Internship Takeaways & Outcome

By anchoring every technical tool in clear analytical questions and rigorous data-quality verification, Pithambar transitioned from foundational chart critique to delivering multi-layered BI dashboards, reproducible Python GIS pipelines, and a full-stack civic environmental platform ready for stakeholder decision-making.

CHAPTER I

INTRODUCTION

1.1 Background & Evolution of Data Visualisation

In modern data-driven organizations and civic governance, raw tabular datasets rarely persuade decision-makers or reveal hidden structural anomalies on their own. Data visualisation bridges the gap between high-volume machine-readable records and human visual cognition. When numerical quantities are encoded through pre-attentive visual channels—such as position along a common scale, length, spatial proximity, or sequential colour saturation—an analyst can detect trends, outliers, and correlations within seconds rather than minutes of row-by-row inspection.

Over the past two decades, the practice of data visualisation has evolved through three distinct waves. The first wave relied on static spreadsheet charts generated inside desktop office suites, where decorative 3D effects, exploded pie slices, and dual-axis clutter often compromised perceptual accuracy. The second wave introduced enterprise Business Intelligence (BI) platforms such as Microsoft Power BI and Tableau, pairing relational Star Schema data models and expression languages like DAX with interactive cross-filtering. The third and current wave combines programmatic Python ecosystems—Pandas, Matplotlib, Seaborn, Plotly, and Streamlit—with web-native mapping frameworks such as Leaflet.js and RESTful APIs like Flask to deliver reproducible, browser-accessible data stories.

Within the Bachelor of Computer Applications (BCA — Data Science) curriculum at MVJ Degree College, Bengaluru, the Final Year Internship (Course Code: DS5P3, Data Visualisation Track) is designed to immerse students in all three waves sequentially. Rather than treating visualisation as a superficial styling step at the end of an analysis, the track emphasizes end-to-end responsibility: formulating the right analytical question, auditing raw sources for missing keys or hidden aggregate rows, choosing perceptually honest chart encodings, and deploying interactive artefacts collaboratively via version control.

> **Core Principle of the Internship**
>
> “Start with the question you want the chart to answer; verify the underlying data integrity first; and select the visual encoding that communicates the truth accurately without unnecessary visual distortion.” — Adapted from Team 1 Project 1 Chart Choice Guidelines.

1.2 Problem Statement & Civic Data Context in Bengaluru

Bengaluru, India's fastest-growing technology metropolis, generates immense volumes of urban, meteorological, transit, demographic, and ecological data every day. However, much of this civic data is published across fragmented portals in inconsistent formats—ranging from multi-sheet Excel workbooks and unnormalized CSV exports to legacy KML geospatial files and census spreadsheets. Without systematic cleaning, schema alignment, and thoughtful visual communication, urban planners, researchers, and citizens struggle to extract actionable insights.

During this internship, Team 1 tackled four concrete problem domains rooted in Bengaluru's real-world civic datasets:

- **1. Misleading Visuals & Climate Variability:** Public charts frequently distort comparisons using 3D perspective cones or crowded multi-series bars. Simultaneously, understanding long-term Bengaluru monsoon behaviour (1902–2025) requires cleaning historical rainfall records and aligning them with El Niño and La Niña climatic flags.
- **2. Urban Transit Congestion (Namma Metro):** Bengaluru Metro Rail Corporation Limited (BMRCL) ridership logs span tens of thousands of hourly station-entry, station-exit, and origin-destination (OD) records. Unclean station codes, blank aggregate summary rows, and wide 24-hour column layouts prevent instant peak-hour and month-over-month reporting until reshaped into a Star Schema.
- **3. Housing Affordability & Ward-Level Civic Equity:** Real estate prices vary sharply across Bengaluru area types, while BBMP ward-level public goods and Census 2011 demographics suffer from inconsistent ward naming across GeoJSON and CSV files, blocking automated spatial joins.
- **4. Urban Lake Degradation & Encroachment:** Bengaluru's 207 lakes form an interconnected stormwater and ecological network, yet custody records (BBMP, BDA, Forest Dept.) and geospatial KML coordinates exist in separate silos, obscuring the link between lake encroachment, water quality (WQI, DO, BOD), and biodiversity collapse.

Addressing these four problem statements required building clean, auditable ETL pipelines and pairing them with domain-appropriate static, BI, and web-interactive visualisations.

1.3 Objectives of the Internship

The primary goal of Pithambar's internship was to master the complete lifecycle of data visualisation—from raw data ingestion and structural cleaning to enterprise BI modelling and interactive full-stack web storytelling. The specific technical and analytical objectives established at the outset were:

- **Master Visual Perception & Chart Selection:** Distinguish between effective and misleading chart designs by evaluating scale integrity, clutter, and cognitive load, and map analytical questions to optimal chart types (bar, line, histogram, scatter, heatmap, violin, choropleth).
- **Develop Rigorous Data Cleaning & ETL Proficiency:** Clean, standardize, and merge heterogeneous multi-sheet Excel and CSV datasets, isolate hidden summary rows, validate totals, and maintain transparent Data Quality audit logs.
- **Build Relational Business Intelligence Models:** Design Star Schema data models in Microsoft Power BI by connecting fact tables to conformed dimension tables, author analytical DAX measures using filter context functions (CALCULATE, MAXX, DIVIDE), and design executive dashboard pages.
- **Create Static & Interactive Python Visualisations:** Utilize Pandas, Matplotlib, Seaborn, and Plotly Express to produce both high-DPI publication figures (300 DPI) and standalone interactive HTML visualisations, alongside prototyping data apps in Streamlit.
- **Perform Geospatial Data Joining & Choropleth Mapping:** Normalize municipal ward identifiers across GeoJSON boundary files and CSV attribute tables to build interactive cartographic maps using Plotly and Leaflet.js.
- **Deliver a Full-Stack Capstone Visualisation Platform:** Contribute to a cohesive Flask + JavaScript + Python analytics platform (*Lakes of Bengaluru*) that integrates automated KML/CSV parsing, REST APIs, cached statistical charts, and before-and-after rejuvenation narratives.
- **Practice Collaborative Software Engineering:** Operate in a three-member team using Git branching/merging, structured documentation in Markdown, sprint reviews, and disciplined daily time-logging.

1.4 Scope, Methodology & Team Collaboration Model

The scope of the internship spanned four sequential projects executed over a 32-day calendar window (17 October to 17 November 2026), encompassing 27 active logged working days (excluding two university exam days on 27–28 October and two Diwali holiday dates on 8–9 November 2026), for a verified total of **92 working hours**.

Iterative Agile-Style Methodology:

Each project followed a four-step engineering cycle: (1) *Source Discovery & Audit*—inspecting schemas, missing values, and data caveats; (2) *Transformation & Modelling*—normalizing keys, unpivoting wide columns, or building relational joins; (3) *Visual Encoding & Refinement*—selecting appropriate colour palettes (e.g., ColorBrewer-inspired safe palettes, diverging RdBu_r for correlations, YlGnBu for spatial density) and direct annotations; and (4) *Mentor Review & Documentation*—presenting outputs during scheduled Google Meet review sessions (Days 9, 19, and 25) and committing clean code and READMEs to GitHub.

Three-Member Team Collaboration Model (Team 1):

Rather than duplicating identical tasks, Team 1—comprising **Vimal**, **Pithambar**, and **Nithin Kumar**—adopted a modular pipeline where each member owned distinct stages and handoffs:

- **Project 1 (Chart Makeover & Rainfall):** Each member critiqued 2 bad and 2 good charts and built 1 Excel makeover; Vimal collected rainfall data, Pithambar cleaned/expanded the historical 1902–2025 workbook, and Nithin ran statistical analysis.
- **Project 2 (Namma Metro):** Vimal collected raw August/September Excel files; Pithambar cleaned, merged, audited, and built Star Schema relationships, 3 DAX measures, and Dashboard Page 1; Nithin built unpivoted analysis sheets, dimension tables, and Page 2.
- **Project 3 (Python, Choropleth & Ward Story):** Members split static and interactive plot types on Bengaluru housing data, built the BBMP public goods choropleth (Pithambar coded the normalization & Plotly choropleth), and divided the Bengaluru Ward Story into Member 1 (Data Prep — Nithin), Member 2 (Plotly Visuals — Pithambar), and Member 3 (Story & Publishing — Vimal).
- **Capstone (Lakes of Bengaluru):** Team 1 collaborated on the one-page charter, Flask backend, KML/CSV ETL pipeline, 6 Matplotlib/Seaborn charts, and Leaflet interactive web platform.

1.5 Internship Roadmap & Deliverable Summary Matrix

Table 1.1 summarizes the chronological progression of Pithambar's internship across all four phases, mapping dates, logged hours, civic datasets, software stack, and verifiable repository deliverables.

Phase	Timeline & Hrs	Primary Datasets	Tools Used	Pithambar's Key Deliverables
Phase 1: Foundations & Excel/SQL	Oct 17 – Oct 25 (32 hrs)	Good/Bad Charts, Chart Choices, Rainfall (1902–2025)	Excel, SQL, GitHub, VS Code	Critiqued 4 charts; built pithambar_chart_make_over.xlsx; modified rainfall dataset
Phase 2: Namma Metro BI & DAX	Oct 26 – Oct 31 (16 hrs)	BMRCL Aug & Sep 2025 Ridership (55.5M rides)	Excel ETL, Power BI, DAX	cleaned_merged_station_ridership.xlsx; Star Schema relationships; 3 DAX measures & Page 1
Phase 3: Python, GIS & Ward Story	Nov 1 – Nov 7 (22 hrs)	Bengaluru House Data, BBMP Goods, Census 2011 Wards	Python, Pandas, Seaborn, Plotly, Streamlit, Tableau	pithambar_tools_learned.md; violin_plot.py; interactive_heatmap.py; chloropleth.py; member2_visualizations.py
Phase 4: Capstone & Report	Nov 10 – Nov 17 (22 hrs)	OpenCity KML, BBMP Custody CSV (207 Lakes)	Flask, Python, Matplotlib, Seaborn, Leaflet, HTML/JS	Capstone charter contribution; backend improvements & Python charts; Interview prep & Final Report

Table 1.1: Four-Phase Internship Roadmap & Pithambar's Core Deliverables

Structure of the Remaining Report:

- **Chapter II (Description of the Organization):** Details the DS5P3 internship track structure, vision/mission, Team 1 workflow architecture, full technology stack, civic data custodians, and review governance.
- **Chapter III (Experiential Learning):** Provides a technical deep dive into every project executed by Pithambar, accompanied by embedded chart outputs, Star Schema diagrams, quantitative audit tables, and debugging lessons.
- **Chapter IV (Internship Outcomes & Conclusion):** Evaluates the 92-hour effort distribution, competency matrix, cross-project urban insights, stakeholder recommendations, and final conclusion.
- **Bibliography & Annexures:** Lists authoritative references and includes the complete 27-day timesheet log plus selected Python and DAX source code listings.

CHAPTER II

DESCRIPTION OF THE ORGANIZATION

2.1 Overview of the DS5P3 Data Visualisation Track & Mentorship Framework

This internship was executed under the structured industry-readiness framework of the **BCA (Data Science) Final Year Internship Program (Course Code: DS5P3 — Data Visualisation Track, Academic Year 2026–27)** coordinated in collaboration with **MVJ Degree College, Bengaluru** (affiliated to Bengaluru North University) and guided by industry mentor **Mr. Ajay Balamurugadas**.

Unlike traditional classroom assignments that rely on pre-cleaned synthetic CSV files, the DS5P3 Data Visualisation Track is organized like a boutique civic-analytics studio. Interns operate in small three-member agile squads (Team 1: Pithambar, Nithin Kumar, and Vimal) connected via GitHub repositories, Visual Studio Code, and scheduled Google Meet sprint reviews. Every sprint mirrors real industry workflows: receiving ambiguous multi-source public datasets, negotiating schema handoffs between team members, enforcing data-quality validation before plotting, and defending design choices against an explicit 100-mark evaluation rubric (Section 14).

The organizational model emphasizes *transferable engineering discipline*. Students are required to maintain individual daily timesheets (logging start/end timestamps, task descriptions, and cumulative hours), commit code regularly to a shared Git repository (`mvj-internship-dataviz-team1`), document tool learnings in Markdown, and deliver a live full-stack capstone web platform capable of running locally or via static fallback.

Program Profile Snapshot

Program: BCA (Data Science) Final Year Internship — DS5P3 Data Visualisation Track
Institution: MVJ Degree College, Bengaluru (Bengaluru North University)
Industry Mentor: Mr. Ajay Balamurugadas | **Squad:** Team 1 (Pithambar, Nithin Kumar, Vimal)
Duration & Intensity: 17 Oct 2026 – 17 Nov 2026 (27 Active Days, 92 Verified Hours)

2.2 Vision, Mission & Pedagogical Philosophy

Vision:

To cultivate industry-ready Data Science graduates who transform raw, fragmented urban and enterprise datasets into truthful, accessible, and decision-oriented visual narratives that serve both business intelligence and civic transparency.

Mission:

- Instill perceptual literacy before tool syntax so interns select encodings based on human cognition rather than default software templates.
- Build rigorous data-engineering habits—auditing duplicates, isolating hidden aggregate rows, and normalizing messy join keys prior to visualisation.
- Provide hands-on mastery across the full analytics spectrum: spreadsheets, relational BI models, programmatic Python plotting, spatial GIS mapping, and web APIs.
- Foster collaborative accountability through clear role division, shared repositories, peer handoffs, and regular mentor critiques.

Pedagogical Philosophy — “Question First, Audit Second, Visualise Third”:

A defining hallmark of the mentorship under Mr. Ajay Balamurugadas is the rejection of arbitrary “chart generation for its own sake.” Beginning on Day 2 and Day 3, interns learned to critique visual distortions—such as 3D perspective skew and truncated baselines—and to write down explicit analytical questions before opening Excel, Power BI, or Python. Equally important is the habit of documenting limitations transparently: for instance, explicitly flagging approximate values reconstructed from chart images, missing line attributes in metro hourly files, or historical 198-ward boundaries versus current 369 Greater Bengaluru Authority (GBA) wards.

2.3 Organizational & Team 1 Role Structure

Within the DS5P3 Data Visualisation Track, organizational governance flows from the Industry Mentor and MVJ Degree College Faculty Coordinators to collaborative three-member project teams. Team 1 adopted a functional pipeline architecture where data acquisition, ETL/modelling, visual generation, and frontend storytelling rotated cleanly across members so every intern gained breadth while owning mission-critical deliverables.

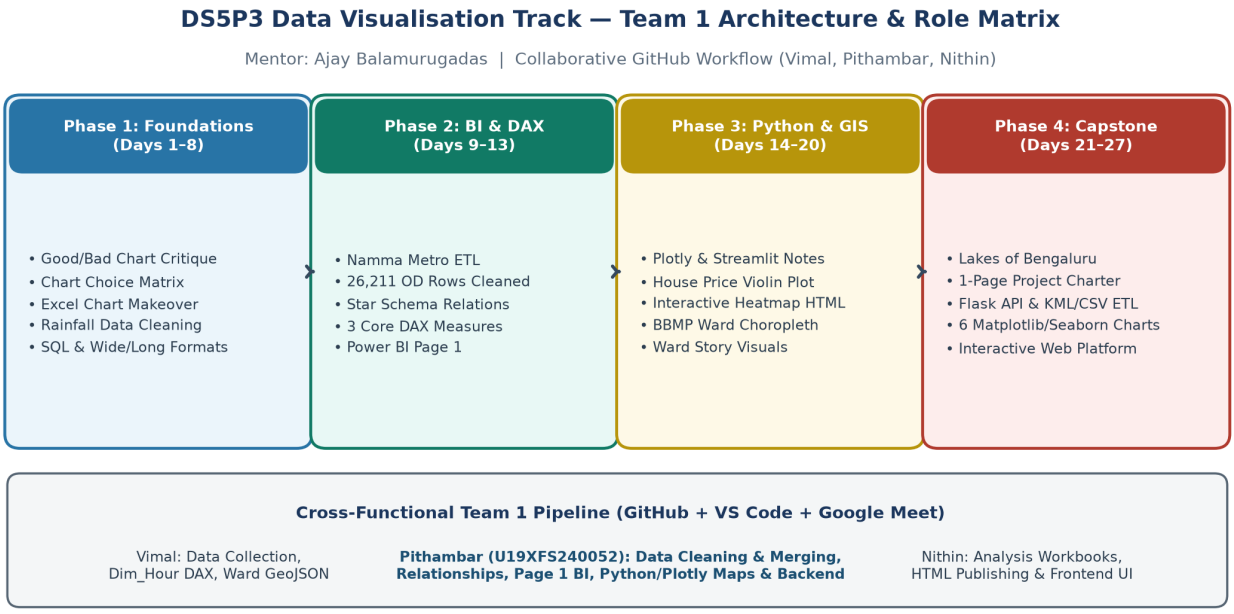


Figure 2.1: DS5P3 Data Visualisation Track — Team 1 Architecture, Four-Phase Progression & Role Matrix

Pithambar's Functional Ownership Within Team 1:

- **Data Cleaning & Integration Lead (Projects 1 & 2):** Modified the 1902–2025 Bengaluru rainfall dataset and engineered the unified *cleaned_merged_station_ridership.xlsx* workbook for Namma Metro.
- **Relational & DAX Modeller (Project 2):** Established Star Schema table relationships in Power BI, authored three peak-hour/average DAX measures, and built Dashboard Page 1.
- **Python & Spatial Visualisation Developer (Project 3 & Capstone):** Built violin plots, interactive Plotly heatmaps, BBMP ward choropleths, the 4-chart Bengaluru Ward Story visual suite, and backend chart/API enhancements for *Lakes of Bengaluru*.

2.4 Technology Stack & Analytical Infrastructure

To mirror real-world analytics teams, the internship deliberately avoided vendor lock-in to a single proprietary tool. Instead, Team 1 integrated open-source Python libraries with industry-standard BI platforms and web technologies. Table 2.1 catalogs the complete software stack utilized across Pithambar's 92 logged hours.

Layer	Technologies & Libraries	Role in Internship Projects
Spreadsheet & BI	Microsoft Excel, Power BI Desktop, DAX, Tableau	Data cleaning, chart makeovers, Star Schema relationships, calculated measures, 3-page interactive dashboards
Programming & Query	Python 3.13, SQL, HTML5, CSS3, JavaScript (ES6)	Automated ETL pipelines, relational querying, backend API routing, responsive SPA frontend
Python Data & Viz	Pandas, NumPy, Matplotlib, Seaborn, Plotly Express, Streamlit	Dataframe reshaping, 300-DPI statistical plots (violin, KDE, regplot, boxplot), interactive HTML charts & web apps
Geospatial & Mapping	GeoJSON, KML (XML), Plotly Choropleth Map, Leaflet.js	Coordinate parsing, NFKD ward-name normalization, Carto Positron & OpenStreetMap interactive maps
Web Framework	Flask (Python REST API), CORS, JSON Data Stores	Serving /api/lakes, /api/stats, /api/story, and dynamic zone-filtered /api/charts PNG endpoints
Collaboration & Dev	Git, GitHub, Visual Studio Code, Google Meet, Markdown	Version control, branch merging, modular repo organization, remote sprint reviews, technical documentation

Table 2.1: Comprehensive Technology Stack & Tooling Infrastructure

A notable architectural highlight of the Capstone infrastructure is its **zero-API-key, dual-mode design**: the *Lakes of Bengaluru* platform serves dynamic Python-generated Matplotlib/Seaborn charts and filtered JSON records via Flask (`http://localhost:5000`), while simultaneously shipping locally hosted Leaflet assets and a `data_fallback.js` layer so the frontend opens cleanly even without a running server.

2.5 Open-Data Ecosystem & Civic Data Custodians

Every analytical project in this internship leveraged authentic civic datasets centered on Bengaluru. Working with public-sector records exposed the team to real-world data imperfections—multi-line cell breaks, embedded summary rows, inconsistent ward spellings, and disparate coordinate systems—that rarely appear in textbook datasets.

Project Domain	Custodian / Source Portal	Scale & Key Schema Attributes
Bengaluru Rainfall (Project 1)	Historical Meteorological Records (1902–2025)	125 rows x 16 cols (monthly Jan–Dec rainfall, annual totals, El Niño / La Niña flags)
Namma Metro Ridership (Project 2)	BMRCL Station-wise Hourly & OD Matrix (Aug–Sep 2025)	55,505,708 total rides; 84 stations; 48 active dates; 26,211 August OD rows
Bengaluru Housing (Project 3)	Bengaluru House Price Dataset (Bengaluru_House_Data.xlsx)	Area type, location, size, total_sqft, bath, balcony, and price (Lakhs INR)
BBMP Public Goods (Project 3)	OpenCity / BBMP Ward-wise Public Goods & GeoJSON	199 CSV rows x 29 civic attributes (population, area, roads, lakes) + BBMP-wards.geojson
Bengaluru Ward Story (Project 3)	DataMeet Spatial Boundaries & OpenCity Census 2011	198 matched BBMP wards; 8,443,675 total population; male/female literacy & worker counts
Lakes of Bengaluru (Capstone)	OpenCity.in KML, BBMP Lakes Custody & KSPCB Benchmarks	207 lakes (5,390.4 acres across 8 zones), WQI, DO, BOD, COD, encroachment %, biodiversity

Table 2.2: Open Civic Data Custodians & Source Datasets Utilized

Through the Capstone project, Team 1 also analyzed institutional governance across Bengaluru's water bodies: **BBMP Lakes Division** stewards approximately 74% of municipal lakes, alongside joint taskforces with the **Bangalore Development Authority (BDA)**, the **Karnataka Forest Department**, and pioneering citizen co-management trusts such as **MAPSAS** (Kaikondrahalli Lake) and the **Jakkur Fisherfolk Cooperative**.

2.6 Quality Assurance, Sprint Reviews & Timesheet Governance

Maintaining analytical trustworthiness across a three-person team required explicit quality gates and structured governance practices throughout the 32-day program:

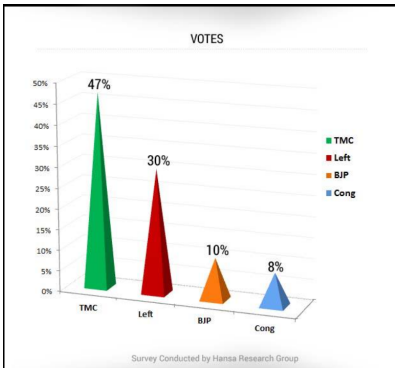
- **1. Dedicated Data Quality Audit Logs:** In Project 2, Pithambar embedded a dedicated *Data Quality* sheet inside *cleaned_merged_station_ridership.xlsx* documenting row counts, blank-station aggregate rows separated (1,305 in the August OD Matrix), duplicate checks (0 duplicates in hourly/entry/exit sheets), and formula recalculations.
- **2. Defensive Validation in Python Scripts:** In Project 3's *chloropleth.py*, Pithambar implemented explicit pre-flight assertions checking file existence, rejecting duplicate normalized ward keys rather than silently summing ambiguous records, and reporting exact matched vs. unmatched row counts.
- **3. Transparent Caveats & Metadata Notes:** Every deliverable documents known limitations—such as estimated values read from pie-chart images in Project 1, absence of a metro-line attribute in Project 2's combined hourly sheet, and the 198-ward historical scope of Census 2011 in Project 3.
- **4. Scheduled Mentor Sprint Reviews:** On Day 9 (25 Oct), Day 19 (6 Nov), and Day 25 (15 Nov), Team 1 convened morning review sessions (08:00–10:00) to demo live workbooks, interactive HTML outputs, and Flask endpoints, incorporating mentor feedback immediately.
- **5. Daily Timesheet & Git Hygiene:** Every active session was logged in *timesheet/pithambar_timesheet.xlsx* with exact dates, start–end windows, deliverables, and hours (92 total hours), synchronized via Git commits to `mvj-internship-dataviz-team1`.

CHAPTER III

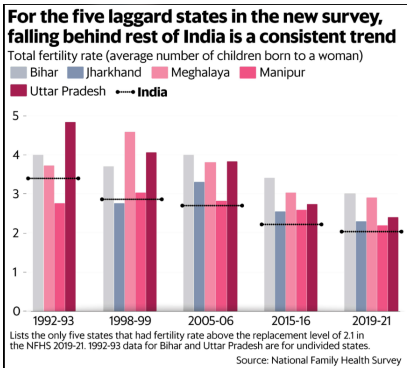
EXPERIENTIAL LEARNING

3.1 Phase 1: Visual Perception, Chart Critique & Chart Choices (Days 1–3)

Following environment setup (GitHub, VS Code, Google Meet, and repository initialization on Day 1), Pithambar focused on visual perception principles. Each team member curated two flawed ("bad") charts and two exemplary ("good") charts. Pithambar's assigned portfolio (Images 3 and 4) contrasted perceptual distortion against direct, uncluttered visual encoding:

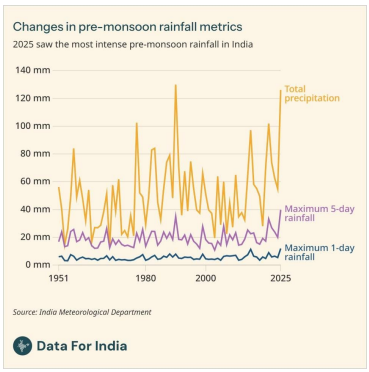


Bad Img 3: 3D Cones distort vote share heights

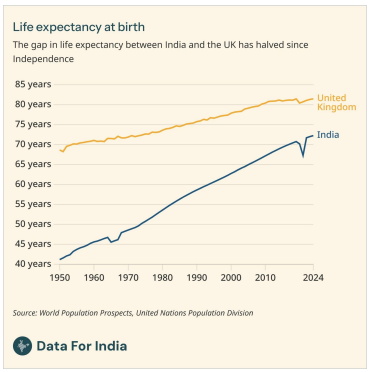


Bad Img 4: Cluttered bars & markers (Fertility)

Figure 3.1: Bad Charts Assigned to Pithambar for Perceptual Critique



Good Img 3: Clean time-series (Pre-monsoon rainfall)



Good Img 4: Direct labels (India vs. UK Life Expectancy)

Figure 3.2: Good Charts Assigned to Pithambar Demonstrating Direct Labelling & Scale Clarity

On Day 3, Team 1 synthesized `project1/chart_choices.md`, mapping eight core analytical questions (trends, rankings, distributions, part-to-whole shares, correlations, and two-dimensional seasonal matrices) to recommended chart types while noting prerequisite data constraints (e.g., plotting rainfall vs. temperature only when both variables exist).

3.2 Phase 1: Excel Chart Makeover & Bengaluru Rainfall ETL (Days 4–8)

On Day 4, Pithambar converted a multi-slice pie chart (*bad chart img5.png*, showing global Country/Region shares) into a clean horizontal bar chart in *project1/chart_make_over/make_over_1/pithambar_chart_make_over.xlsx*. Pie charts encode quantities via angles and arc areas, making it difficult to distinguish similarly sized categories such as *Other (31.5%)* vs. *China (29.5%)* or *India (6%)* vs. *Japan (5%)* vs. *Russia (4%)*. Replacing radial wedges with sorted horizontal bars anchored to a common zero baseline made magnitude comparisons instantaneous. Pithambar also documented a data transparency note clarifying that the plotted percentages were approximations read from the original image.

Country / Region	Estimated Share (%)	Perceptual Advantage of Bar Redesign
Other	31.5%	Top composite category; effortless rank reading on horizontal bar
China	29.5%	Nearly equal to 'Other' (2% gap invisible in pie, obvious in bar)
U.S.	14.0%	Clearly visible as half the size of China
EEA	10.0%	Ranked 4th with exact 10% baseline alignment
India	6.0%	Clean comparison against Japan and Russia
Japan	5.0%	1% difference over Russia preserved accurately
Russia	4.0%	Smallest slice, clearly readable without crowded pie callouts

Table 3.1: Pithambar's Chart Makeover Data Table (Pie Chart Converted to Horizontal Bar)

Bengaluru Rainfall Dataset Modification & Dashboard Support (Days 5–8):

While Vimal collected recent 2024–2025 rainfall records (*banglore_rainfall_data.xlsx*), Pithambar cleaned and expanded the dataset into *modified_banglore_rainfall_data.xlsx* covering a long-term historical series of **125 rows and 16 columns** spanning **1902 to 2025** (monthly values Jan–Dec, annual Total, and binary *El NiNo (Y/N)* and *La Nina (Y/N)* indicators). Pithambar also spent Day 6 learning SQL querying concepts, Day 7 organizing wide-vs-long data structures, and Day 8 building before-and-after Excel dashboards.

3.3 Phase 2: Namma Metro Ridership Data Cleaning & Merging (Days 9–10)

Following the Day 9 sprint review, Project 2 transitioned Team 1 to large-scale urban transit analytics using Namma Metro (BMRCL) ridership logs for August and September 2025. Raw files collected by Vimal arrived in three heterogeneous layouts: an August hourly workbook, a two-sheet September entry/exit workbook, and a large August Origin-Destination (OD) matrix.

On Day 10 (26 October 2026), Pithambar engineered the master prepared workbook *project2/namma metro/pithambar data cleaning and merging/cleaned_merged_station_ridership.xlsx*, consolidating all sources into five structured sheets:

- **Combined Hourly (6,336 rows × 30 cols):** Unified August and September station records with standardized *source*, *measure* (Entry/Exit), *date*, *station*, 24 hourly slots, and recalculated *TOTAL*.
- **Station Lookup (167 rows × 2 cols):** Extracted and mapped canonical *STATION CODE* to *STATION NAME* pairs (e.g., 11 → Baiyappanahalli, 110 → Kempegowda, 111 → KSR Railway Station).
- **August OD Matrix (26,212 rows × 88 cols) & August OD Aggregates (1,306 rows × 87 cols):** Isolated 1,305 blank-station summary rows from granular station-to-station rows, preventing double-counting in downstream aggregations.
- **Data Quality (5 rows × 5 cols):** Embedded an auditable data-quality scorecard (reproduced in Table 3.2).

Source Dataset	Station Records	Blank Aggregates	Duplicates	Cleaning Actions Performed by Pithambar
August 2025 hourly	1,355	0	0	Hourly values standardized; TOTAL recalculated from 24 hourly columns.
September 2025 entry	1,245	0	0	Station names split into code & name; TOTAL recalculated.
September 2025 exit	1,245	0	0	Station names split into code & name; TOTAL recalculated.
August 2025 OD matrix	26,211	1,305	Not evaluated	Blank-station aggregate rows separated from station records; date/time parsed.

Table 3.2: Namma Metro Ridership Data Quality Audit (`Data Quality` Sheet)

3.4 Phase 2: Power BI Star Schema, DAX Measures & Page 1 Dashboard (Days 11–13)

After completing university exams on 27–28 October, Pithambar resumed on Day 11 to build the relational **Star Schema** inside Power BI (*star schema.pbix*), linking the central 152,040-row `Fact_Ridership` table (84 stations, 48 active dates from Aug 1 to Sep 30, 2025) to conformed dimension tables (`Dim_Station`, `Dim_Date`, and Vimal's DAX-generated `Dim_Hour`).

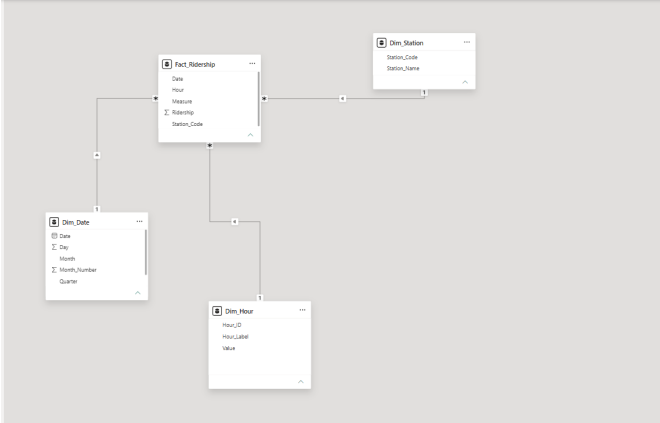


Figure 3.3: Namma Metro Power BI Star Schema Relationships Established by Pithambar

On Days 12–13, Pithambar authored and validated three core DAX measures—*Average Ridership*, *Peak Hour Ridership*, and *Peak Hour Share* (ensuring Percentage formatting without multiplying by 100)—and designed **Page 1** of *namma metro power bi dashboard/dashboard.pbix*. Table 3.3 summarizes all ten team KPIs verified in *final report of all 10 measures.csv*.

Measure Name	Owner	Reported Value	DAX Formula
Total Ridership	Nithin	55,505,708	SUM(Fact_Ridership[Ridership])
Total Entries	Nithin	21,721,271	CALCULATE([Total Ridership], Fact_Ridership[Measure] = "Entry")
Total Exits	Nithin	21,667,826	CALCULATE([Total Ridership], Fact_Ridership[Measure] = "Exit")
Average Ridership	**Pithambar**	**365.073**	AVERAGE(Fact_Ridership[Ridership])
Peak Hour Ridership	**Pithambar**	**5,376,931**	MAXX(ALL(Dim_Hour[Value]), CALCULATE([Total Ridership]))
Peak Hour Share	**Pithambar**	**9.69% (0.0969)**	DIVIDE([Peak Hour Ridership], [Total Ridership], 0)
Weekday Ridership	Vimal	42,530,867	CALCULATE([Total Ridership], Dim_Date[Day Type] = "Weekday")
Weekend Ridership	Vimal	12,974,841	CALCULATE([Total Ridership], Dim_Date[Day Type] = "Weekend")
Previous Month Ridership	Vimal	12,116,611	CALCULATE([Total Ridership], DATEADD(Dim_Date[Date], -1, MONTH))
MoM % Change	Vimal	3.581 (358.1%)	DIVIDE([Total Ridership]-[Previous Month Ridership], [Previous Month Ridership], 0)

Table 3.3: Namma Metro Power BI 10-Measure KPI Snapshot (Highlighting Pithambar's 3 Measures)

3.5 Phase 3: Python Modules (Plotly, Streamlit) & House Price Violin Plot (Days 14–15)

Entering Phase 3 on 1 November 2026 (Day 14), Team 1 shifted from desktop BI tools to programmatic Python data visualisation. Pithambar authored *project3/learning tools/pithambar_tools_learned.md*, documenting practical patterns for combining **Plotly Express** (`px.bar`, hover inspection, interactive zooming) with **Streamlit** (`st.title`, `st.text_input`, `st.bar_chart`, `streamlit run app.py`) to convert standalone Python scripts into browser-based analytical web applications.

On Day 15 (2 November 2026), Pithambar built *project3/python plots/violin_plot.py* using Pandas, Matplotlib, and Seaborn to analyze *Bengaluru_House_Data.xlsx*. After coercing the `price` column to numeric values and dropping missing observations across `area_type` and `price`, he rendered a high-resolution (300 DPI) violin plot with internal quartile markers (`inner='quartile'`, `cut=0`) to reveal both median price differences and full probability density tails across Super built-up, Built-up, Plot, and Carpet area categories.

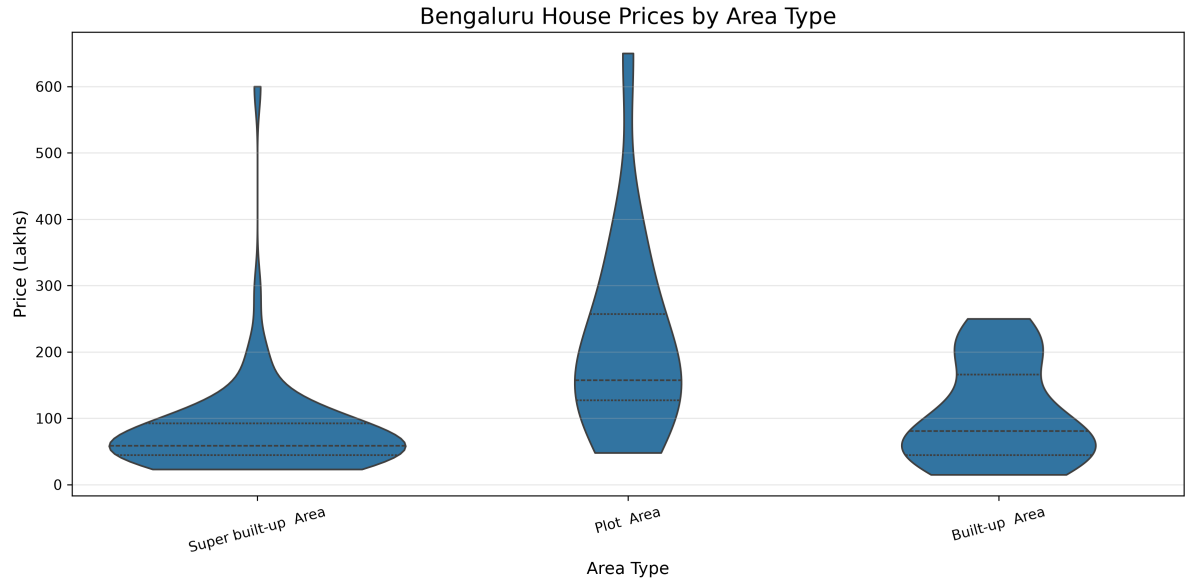


Figure 3.4: Bengaluru House Prices by Area Type — Seaborn Violin Plot Created by Pithambar (`violin_plot.py`)

> **Analytical Observation from the Violin Plot**
>
> Unlike a standard bar chart of mean prices, the violin plot exposes heavy right-skewness across *Plot Area* and *Super built-up Area* properties, showing that luxury outliers inflate mean prices far above the inner quartile medians.

3.6 Phase 3: Interactive Correlation Heatmap & Exploratory Plots (Day 16)

Building directly on static exploratory analysis, Pithambar transitioned on Day 16 (3 November 2026) to browser-ready interactive plotting with Plotly. In *project3/interactive plots/interactive_heatmap.py*, he filtered *Bengaluru_House_Data.xlsx* dynamically for numerical attributes (`select_dtypes(include='number')`), computed the Pearson correlation matrix (`numeric_df.corr()`), and rendered an interactive matrix via `px.imshow()`.

Crucial perceptual design choices included locking the colour scale strictly to `zmin=-1` and `zmax=1`, applying a diverging `RdBu_r` palette so positive and negative correlations remain visually distinct around zero, enabling two-decimal cell annotations (`text_auto='.2f'`), and exporting the self-contained visual to *bengaluru_interactive_heatmap.html* for zero-dependency browser exploration alongside team boxplots, histograms, scatter plots, and animated time charts.

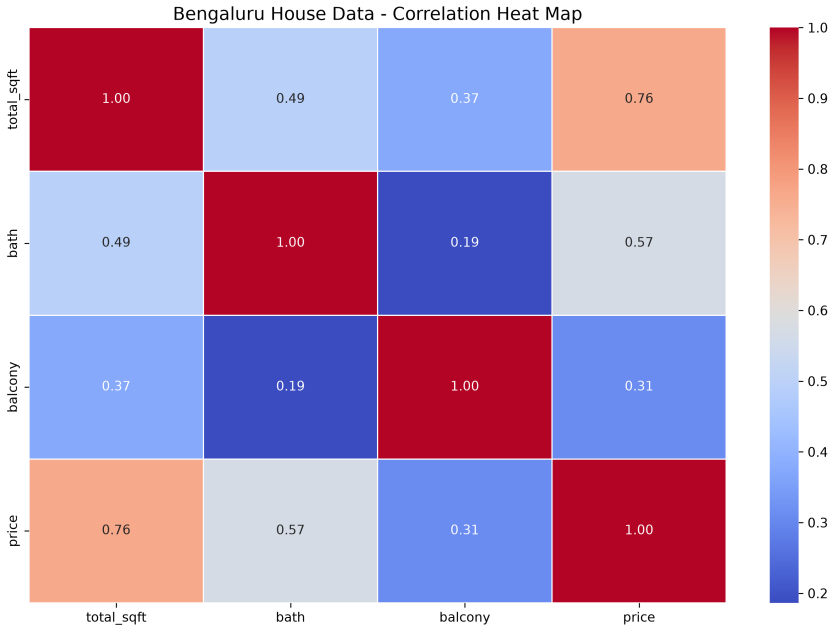


Figure 3.5: Bengaluru House Data Correlation Heatmap Reference (Numerical Feature Relationships)

Inspecting the correlation matrix confirms strong positive correlation between bathroom count (`bath`) and house `price`, moderate correlation with `balcony`, and near-zero correlation with synthetic index identifiers—guiding which features hold genuine predictive signal for downstream housing models.

3.7 Phase 3: BBMP Ward Choropleth, Tableau & Bengaluru Ward Story (Days 17–20)

Between 4 November and 7 November 2026 (Days 17–20), Pithambar specialized in municipal geospatial visualisation. On Day 17, he engineered *project3/BBMP chloropleth/chloropleth.py* to map Vimal's BBMP ward-wise public goods dataset against *BBMP-wards.geojson*. Because civic CSVs and GeoJSON files use inconsistent capitalization, punctuation, and the word "Ward", Pithambar wrote a `normalize()` function using Unicode `NFKD` normalization and regex stripping, auto-detected the highest-overlap property pair, guarded against duplicate ward keys, and generated an interactive Plotly map (`px.choropleth_map`, Carto Positron basemap, centered on Bengaluru `12.9716° N, 77.5946° E`).

On Day 18 (5 Nov), Pithambar installed Tableau and explored its drag-and-drop shelf paradigm on the same BBMP ward workbook (`project3/tableau/`). Following the Day 19 review, Day 20 culminated in the collaborative **Bengaluru Ward Story** (`project3/bangalore ward story telling/`), where Pithambar served as **Member 2** (*member2_visualizations.py*), building four linked Plotly visualisations consumed by Vimal's published `index.html`:

- **1. Interactive Ward Population Choropleth:** `YlOrRd` sequential shading across all 198 historical BBMP wards.
- **2. Top-10 Most Populated Wards Bar Chart:** Horizontal ranking led by Horamavu (95,368), Bellanduru (80,180), and Begur (80,037).
- **3. Grouped Male vs. Female Population Chart & 4. Gender-Balance Scatter Plot:** Comparing total ward size against females per 1,000 males.

Project Component	Dataset Scope	Key Engineering & Analytical Findings
BBMP Public Goods Choropleth	199 CSV rows × 29 cols + BBMP-wards.geojson	NFKD Unicode & regex normalization matching `_ward_key`; duplicate key guard; `YlGnBu` Carto Positron map
Bengaluru Ward Story (Census 2011)	198 matched historical BBMP wards (8,443,675 pop.)	4 Plotly figures in `visualizations.html`: Choropleth, Top-10 Bar, Male/Female Grouped Bar, Gender Scatter
Extreme Benchmarks (Ward Story)	Median ward pop.: 39,551 Median sex ratio: 936.3	Largest: Horamavu (95,368); Smallest: Marenahalli (21,171); Gender ratio: Vasanth Nagar (1,066.8) vs. Agaram (736.5)

Table 3.4: BBMP Ward Public Goods & Census 2011 Spatial Dataset Summary

3.8 Phase 4 (Capstone): Lakes of Bengaluru — Architecture & Data Pipeline (Days 21–22)

After the Diwali break (8–9 November), Team 1 launched the capstone project—**Lakes of Bengaluru (Option 3: Environmental Data Visualisation & Spatial Storytelling)**—spanning Days 21–24. On Day 21, Pithambar contributed to the one-page project charter defining scope, target metrics, and the 100-mark Section 14 evaluation targets. On Days 22–23, he collaborated on website planning and backend improvements.

The platform implements a clean three-tier architecture (`capstone - lakes of bengaluru/`): (1) a **Flask Python Backend** (`app.py`) exposing REST endpoints (`/api/lakes`, `/api/stats`, `/api/story`, `/api/charts/<id>`) with multi-criteria filtering by zone, status, acreage, WQI, and walking-track availability; (2) an automated **ETL Pipeline** (`backend/data_pipeline.py`) parsing OpenCity `bbmp_lakes.kml` coordinates (validating Bengaluru's bounding box `12.5–13.5° N, 77.0–78.0° E`), normalizing BBMP custody CSV zones and acre-gunta extents (`acres + guntas/40`), and outputting `lakes_master.json`, `lakes_master.csv`, and `before_after_stories.json`; and (3) a **Single-Page Frontend** featuring Leaflet.js maps, before/after split sliders, and a searchable 207-lake directory.

Capstone Metric	Verified Value	Analytical Context
Total Water Bodies Mapped	207 Lakes	Unified across OpenCity KML placemarks & BBMP custody CSVs
Total Surface Basin Extent	5,390.4 Acres	Largest outlier: Bellandur Lake (913.0 acres, Mahadevapura)
Developed / Rejuvenated	107 Lakes (51.7%)	WQI 72–92; fenced perimeters, walking trails & STPs/wetlands
Work in Progress	37 Lakes (17.9%)	WQI 50–68; active desilting, bunding, or NGT-monitored works
To Be Developed	46 Lakes (22.2%)	WQI 35–52; priority candidates for sewage diversion & restoration
Degraded / Encroached	17 Lakes (8.2%)	WQI 18–34; severe organic pollution (BOD 50–120 mg/L) & encroachment
Network Average WQI	64.3 / 100	Across 8 BBMP civic zones (Mahadevapura hosts 87 lakes, Bommanahalli 45)

Table 3.5: Lakes of Bengaluru — Master KPI Summary Verified from `lakes_master.json` *(207 Water Bodies)*

3.9 Phase 4 (Capstone): Python Chart Engine & Rejuvenation Case Studies (Days 23–24)

On Day 24 (13 November 2026), Pithambar focused on Python chart generation (`backend/chart_generator.py`), producing six publication-grade Matplotlib and Seaborn charts served both statically and dynamically (supporting `?zone=` filtering via `io.BytesIO` buffers). Four flagship charts below demonstrate the ecological story:

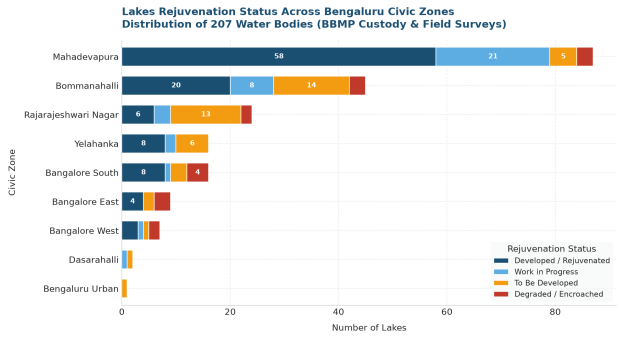


Fig 3.6: Status Breakdown Across Civic Zones

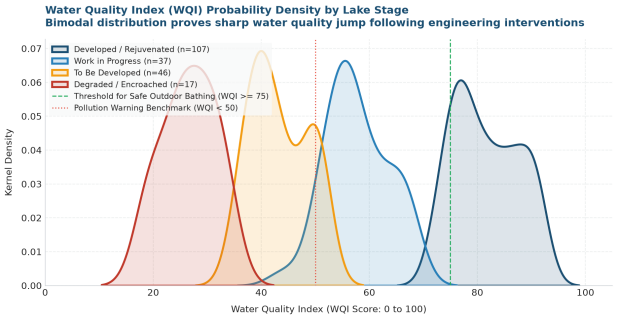


Fig 3.7: Bimodal WQI Density (Rejuvenated vs. Degraded)

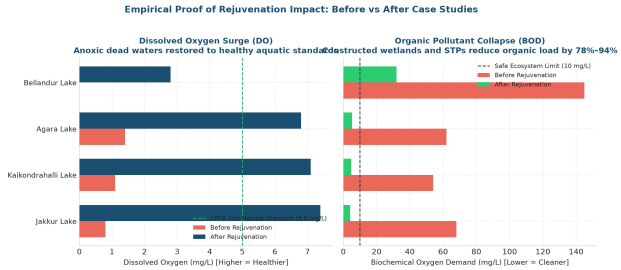


Fig 3.8: DO Surge & 78%–94% BOD Collapse

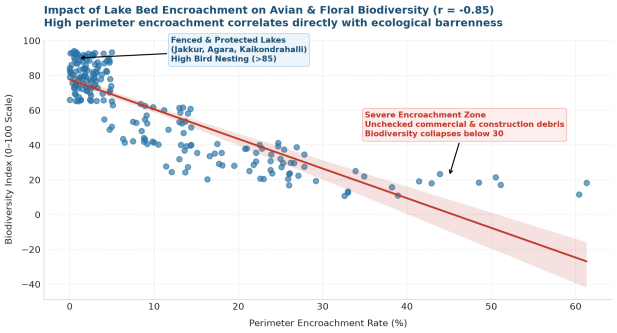


Fig 3.9: Encroachment vs. Biodiversity (r = -0.84)

The four before-and-after case studies—**Jakkur Lake** (160 acres, DO lifted from 0.8 to 7.4 mg/L, 204 bird species), **Kaikondrahalli Lake** (48.2 acres, MAPSAS citizen stewardship, 122 bird species), **Agara Lake** (98 acres, BOD cut from 62 to 5.2 mg/L), and **Bellandur Lake** (913 acres, 3.2M cu.m toxic sludge evacuated, foam events dropped from 48/yr to 0)—quantify how constructed wetlands and sewage diversion reverse ecological collapse.

3.10 Technical Challenges Faced, Debugging & Key Lessons Learned

Transitioning from classroom exercises to real civic datasets confronted Pithambar with non-trivial engineering and analytical bugs. Resolving each issue strengthened his debugging methodology:

- **1. Hidden Aggregate Rows Double-Counting Ridership (Project 2):** Raw August OD Matrix files mixed 1,305 blank-station summary rows directly into 24,906 station-level records. *Resolution:* Separated blank-station rows into a dedicated `August OD Aggregates` sheet and recorded the exact count in `Data Quality` before building Power BI relationships.
- **2. Mismatched Ward Strings Across GeoJSON & CSV (Project 3):** Direct string joins failed due to prefixes ("Ward 12"), trailing spaces, accents, and punctuation. *Resolution:* Implemented `NFKD` Unicode normalization + regex word stripping (`_ward_key`) and built an overlap-scoring auto-matcher plus a strict duplicate-row exception guard.
- **3. Plotly Map API Version Drift (Project 3):** Newer Plotly versions deprecate `px.choropleth_mapbox` in favor of `px.choropleth_map`. *Resolution:* Used `hasattr(px, 'choropleth_map')` runtime feature detection in `member2_visualizations.py` so the script executes cleanly across environments.
- **4. Extreme Area Outliers Compressing Boxplots (Capstone):** Bellandur Lake's 913-acre extent compressed all other 206 lakes (median ~24 acres) against the y-axis. *Resolution:* Capped the x-axis at 170 acres in `chart2_area_distribution.png` and added a high-contrast direct callout arrow annotating Bellandur's true extent.
- **5. Multi-Developer Git Merge Coordination:** Concurrent pushes across three members occasionally caused merge conflicts on shared workbooks and lock files (`~$`). *Resolution:* Enforced clean Git pull-before-push habits and modular per-member folders.

> **Key Lesson Learned**
>
> Roughly 60% of reliable data visualisation happens before a single chart is drawn—inside schema validation, outlier inspection, and key normalization. Once data integrity is guaranteed, clean encodings tell the story effortlessly.

CHAPTER IV

INTERNSHIP OUTCOMES AND CONCLUSION

4.1 Quantitative Internship Outcomes & Effort Analysis

Across the internship period from **17 October 2026 to 17 November 2026**, Pithambar completed all planned milestones across four progressive projects, logging **92 verified hours across 27 active days** (with zero unexplained absences; exam days on Oct 27–28 and Diwali holidays on Nov 8–9 were formally recorded). Figure 4.1 visualizes the distribution of his 92 hours across internship phases and technical competency pillars.

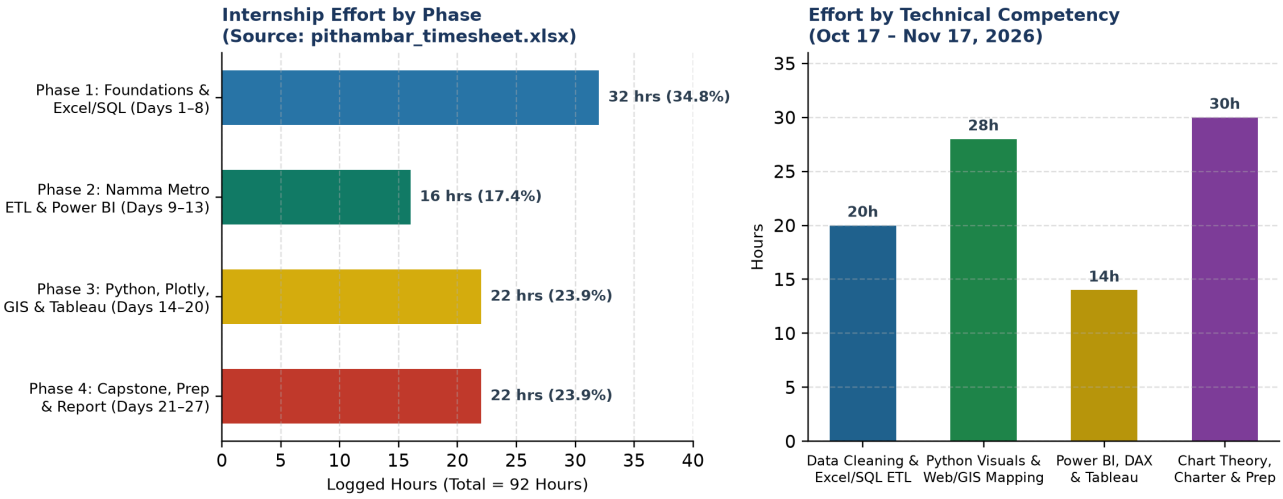


Figure 4.1: Pithambar's 92-Hour Effort Breakdown by Internship Phase & Competency Area (`pithambar_timesheet.xlsx`)

Concrete Quantifiable Deliverables Produced:

- **Datasets Cleaned & Engineered:** 125-row rainfall dataset (1902–2025); 6,336-row unified Metro hourly sheet + 26,211-row OD Matrix; 207-lake JSON/CSV master repository.
- **BI & Analytical Models:** 1 Power BI Star Schema relationship model, 3 validated DAX measures (`365.073` avg, `5.37M` peak hour, `9.69%` peak share), 1 Power BI report page, and 1 Tableau exploration.
- **Python & Web Visuals:** 5+ Python scripts (`violin_plot.py`, `interactive_heatmap.py`, `chloropleth.py`, `member2_visualizations.py`, `chart_generator.py`), generating 11+ static/interactive visual artefacts and supporting the Flask + Leaflet capstone platform.

4.2 Technical & Professional Competencies Acquired

The DS5P3 track transformed theoretical classroom knowledge from Pithambar's BCA (Data Science) coursework into repeatable production habits. Table 4.1 maps the progression of his technical and workplace competencies before and after the 92-hour internship.

Competency Domain	Skill Progression	Verified Evidence from Internship Work
Visual Perception & Design Critique	Beginner → Proficient	Identified 3D distortion & overplotting; built Question-to-Chart matrix & pie-to-bar makeover
Data Cleaning & Spreadsheet ETL	Intermediate → Advanced	Standardized 1902–2025 rainfall; split Metro station codes, separated 1,305 aggregate rows, wrote audit sheets
Power BI & DAX Data Modelling	Beginner → Proficient	Constructed Star Schema joins; authored CALCULATE/MAXX/DIVIDE measures; built dashboard Page 1
Python Plotting (Matplotlib/Seaborn/Plotly)	Beginner → Advanced	Built 300-DPI violin & KDE/regplot visuals, interactive HTML heatmaps, and BytesIO dynamic chart endpoints
Geospatial GIS (GeoJSON & KML)	Novice → Proficient	Parsed OpenCity KML coordinates; wrote NFKD ward-key normalization; built Plotly & Leaflet choropleths
Full-Stack & Git Collaboration	Beginner → Proficient	Improved Flask REST backend; prototyped Streamlit notes; executed 27-day agile Git/sprint workflow

Table 4.1: Technical & Professional Competency Growth Matrix (Pithambar — U19XFS240052)

Soft Skills & Workplace Readiness:

- **Technical Communication:** Presented findings clearly during three formal mentor reviews and interview-prep sessions (Day 26), articulating why specific encodings or DAX filter contexts were chosen.
- **Documentation Discipline:** Authored clear Markdown learning logs (`pithambar_tools_learned.md`), README contributions, and embedded Data Quality notes.
- **Accountability & Time Management:** Balanced academic examinations (Oct 27–28) alongside consistent 4-hour daily engineering blocks to deliver all project handoffs on schedule.

4.3 Key Analytical Findings Across Bengaluru Civic Datasets

Synthesizing results across all four Bengaluru-centric projects uncovered meaningful real-world patterns:

- **1. Long-Term Monsoon Resilience & Climate Signals (Project 1):** Across 1902–2025, Bengaluru's annual rainfall consistently clusters around ~900–1,300 mm (e.g., 1,020.2 mm in 2023, 1,154.0 mm in 2024, 1,045.0 mm in 2025), with clear pre-monsoon and monsoon seasonality that demands multi-decade baselines rather than single-year snapshots.
- **2. Extreme Commuter Concentration in Namma Metro (Project 2):** Across 55,505,708 total rides (84 stations, Aug–Sep 2025), weekdays accounted for 42.53M rides (76.6%) versus 12.97M on weekends. Notably, a single peak hour captured 5,376,931 rides—representing **9.69%** of total network ridership—underscoring acute rush-hour capacity pressure.
- **3. Housing Skewness & Ward-Level Population Disparity (Project 3):** Bengaluru house prices exhibit pronounced right-skewness driven by Plot and Super built-up outliers, correlating strongly with bathroom count. Spatially, across 198 historical BBMP wards (8.44M Census 2011 residents), population ranges over 4.5× from Marenahalli (21,171) to peripheral Horamavu (95,368), while gender balance varies from 736.5 to 1,066.8 females per 1,000 males.
- **4. Quantifiable Ecological Payoff of Lake Rejuvenation (Capstone):** Across 207 Bengaluru lakes (5,390.4 acres), 51.7% (107 lakes) are developed/rejuvenated, producing a sharp bimodal split in Water Quality Index (72–92 vs. <45 for untreated basins). Constructed wetlands and STPs slashed BOD by **78%–94%** and raised Dissolved Oxygen up to 7.4 mg/L, while perimeter encroachment showed a strong negative correlation (**r = -0.84**) with biodiversity.

4.4 Recommendations for Civic Stakeholders & Future Enhancements

Based on the empirical evidence and technical limitations observed during the internship, Pithambar proposes the following actionable recommendations for municipal agencies and future student cohorts:

For Civic & Transit Agencies (BMRCL & BBMP):

- **Publish Standardized Schemas with Explicit Line & Ward IDs:** Namma Metro hourly exports should include a `Metro_Line` (Purple/Green/Yellow) column and exclude embedded blank-station summary rows so BI ingestion requires minimal manual splitting.
- **Prioritize the 63 Unrejuvenated/Degraded Lakes (30.4%):** Target the 46 'To Be Developed' and 17 'Degraded / Encroached' water bodies—especially high-catchment Mahadevapura and Bommanahalli basins—with perimeter bio-fencing and decentralized STP + constructed wetland models proven at Jakkur and Kaikondrahalli.
- **Transition to Citizen Co-Management Trusts:** Replicate the BBMP–MAPSAS partnership model across newly desilted lakes to sustain long-term maintenance beyond initial contractor tenders.

For Future Technical Enhancements of Team 1 Platforms:

- **Migrate Ward Maps to Current 369 GBA Boundaries:** Update Project 3's historical 198-ward GeoJSON and Census 2011 counts once updated Greater Bengaluru Authority spatial files and current demographic projections are published.
- **Automate Live Sensor & Spatial Database Ingestion:** Replace local JSON/CSV snapshots in the *Lakes of Bengaluru* Flask backend with a PostGIS spatial database connected to periodic KSPCB water-quality sensor feeds.
- **Deploy Interactive Cloud Hosting:** Host the Streamlit prototypes and Capstone Flask dashboard on a cloud PaaS so municipal stakeholders can query ward and lake KPIs live.

4.5 Conclusion

The one-month internship under the **DS5P3 Data Visualisation Track** at **MVJ Degree College, Bengaluru** (17 October – 17 November 2026) provided Pithambar (USN: U19XFS240052) with an intensive, authentic bridge between undergraduate Data Science theory and real-world urban analytics engineering.

Over **92 logged hours**, Pithambar progressed methodically from critiquing misleading 3D charts and redesigning spreadsheet visuals to auditing 26,000+ metro origin-destination records, building Star Schema relationships and DAX measures in Power BI, authoring Python violin plots, interactive Plotly heatmaps, and ward-level choropleths, and co-developing the full-stack *Lakes of Bengaluru* environmental data platform.

Equally valuable was the collaborative discipline cultivated inside Team 1 alongside Nithin Kumar and Vimal under the mentorship of Mr. Ajay Balamurugadas. Learning how to document data caveats honestly, write defensive normalization code, synchronize multi-contributor GitHub repositories, and connect quantitative charts to compelling human stories has equipped Pithambar with a strong portfolio and a solid professional foundation for a career in Data Analytics, Business Intelligence, and Data Visualisation.

Final Internship Verification Summary

Student: Pithambar | **USN:** U19XFS240052 | **College:** MVJ Degree College, Bengaluru
Track: BCA (Data Science) — DS5P3 Data Visualisation | **Total Effort:** 92 Hours (27 Active Days)
Repository: mvj-internship-dataviz-team1 (Projects 1, 2, 3 & Capstone: Lakes of Bengaluru)

REFERENCES

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ANNEXURES

ANNEXURE A: INTERNSHIP DAILY TIMESHEET LOG (PART 1)

Source: *timesheet/pithambar_timesheet.xlsx* — Verified record of Pithambar's daily attendance, session timings, tasks completed, and cumulative hours (Days 1–14 shown below; continued on Page 37).

Date	Day	Time Slot	Work Done / Milestone	Hrs
2026-10-17	1	15:00 - 19:00	Set up GitHub,vscode, google meet , and the internship repository.	4
2026-10-18	2	17:00 - 21:00	researched two good and bad charts and pushed it in github, and gave information	4
2026-10-19	3	17:00 - 21:00	collected 3 questions and created a chart choices	4
2026-10-20	4	17:00 - 21:00	remade 1 bad chart into good chart makeover	4
2026-10-21	5	17:00 - 21:00	cleaned banglore rainfall data and made dashboard	4
2026-10-22	6	18:00 - 22:00	Learning SQL	4
2026-10-23	7	16:00 - 20:00	(Data / Exercise Practice)	4
2026-10-24	8	17:00 - 21:00	created 2 before after chart make over with excel dashboard	4
2026-10-25	9	08:00 - 10:00	review	0
2026-10-26	10	11:00 - 15:00	cleaning and merging namma metro raidership	4
2026-10-27	—	—	exam	0
2026-10-28	—	—	exam	0
2026-10-29	11	13:00 - 17:00	power bi creating relationship for star schema	4
2026-10-30	12	14:00 - 18:00	understanding DAX and implementing it in power bi	4
2026-10-31	13	09:00 - 13:00	Page 1 namma metro power bi dashboard.	4
2026-11-01	14	13:00 - 17:00	learned two python modules plotly and streamlit	4

Table A.1 (Part 1): Pithambar's Daily Internship Timesheet Log (Oct 17 – Nov 1, 2026)

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ANNEXURE A: INTERNSHIP DAILY TIMESHEET LOG (PART 2)

Continuation of *timesheet/pithambar_timesheet.xlsx* (Nov 2 – Nov 17, 2026), confirming 27 active logged days and a verified cumulative total of **92 hours**.

Date	Day	Time Slot	Work Done / Milestone	Hrs
2026-11-02	15	09:00 - 13:00	created violion plot using python for banglore house data	4
2026-11-03	16	09:00 - 13:00	created interactive heatmap in python and exported it into html file	4
2026-11-04	17	17:00 - 21:00	created chloropleth map for BBMP ward wise public goods	4
2026-11-05	18	09:00 - 13:00	installed tableau and learned about it	2
2026-11-06	19	08:00 - 10:00	review	0
2026-11-07	20	09:00 - 13:00	Bengaluru ward story created visualization using python	4
2026-11-08	—	—	Diwali Holiday	0
2026-11-09	—	—	Diwali Holiday	0
2026-11-10	21	17:00 - 21:00	helped wth creation of one page capstone charter	4
2026-11-11	22	17:00 - 21:00	creating website for lakes of bengaluru planning	4
2026-11-12	23	09:00 - 13:00	backend improving	4
2026-11-13	24	09:00 - 13:00	using python modules for creating charts and graphs for website	4
2026-11-14	—	—	(Data / Exercise Practice)	0
2026-11-15	25	08:00 - 10:00	review	0
2026-11-16	26	17:00 - 19:00	interview preparation	2
2026-11-17	27	13:00 - 17:00	Report	4
			TOTAL LOGGED HOURS	**92**

Table A.1 (Part 2): Pithambar's Daily Internship Timesheet Log (Nov 2 – Nov 17, 2026)

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ANNEXURE B: PYTHON SOURCE CODE — VIOLIN PLOT & HEATMAP

B.1 Bengaluru House Price Violin Plot (`project3/python plots/violin_plot.py`)

```
# ========================================
# BENGALURU HOUSE PRICE - VIOLIN PLOT
# ========================================
# 1. Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pathlib import Path
# 2. Read the Excel file in the same folder as this script
data_file = Path(__file__).resolve().with_name("Bengaluru_House_Data.xlsx")
df = pd.read_excel(data_file)
# 4. Display dataset
print("First five rows:")
print(df.head())
print("\nDataset Information:")
print(df.info())
# 5. Clean the required columns
df['price'] = pd.to_numeric(df['price'], errors='coerce')
plot_df = df[['area_type', 'price']].dropna()
# 6. Create Violin Plot
plt.figure(figsize=(12, 6))
sns.violinplot(
    data=plot_df,
    x='area_type',
    y='price',
    inner='quartile',
    cut=0
)
# 7. Add chart labels
plt.title(
    'Bengaluru House Prices by Area Type',
    fontsize=16
)
plt.xlabel('Area Type', fontsize=12)
plt.ylabel('Price (Lakhs)', fontsize=12)
plt.xticks(rotation=15)
plt.grid(axis='y', alpha=0.3)
# 8. Save and display chart
plt.tight_layout()
output_file = Path(__file__).resolve().with_name('bengaluru_violin_plot.png')
plt.savefig(
    output_file,
    dpi=300,
    bbox_inches='tight'
)
plt.show()
```

B.2 Interactive Correlation Heatmap (`project3/interactive plots/interactive_heatmap.py`)

```
# 1. Import libraries
import pandas as pd
import plotly.express as px
# 2. Read Excel file
df = pd.read_excel("Bengaluru_House_Data.xlsx")
# 3. Select numerical columns
numeric_df = df.select_dtypes(include="number")
# 4. Calculate correlation
correlation = numeric_df.corr()
# 5. Create interactive heatmap
fig = px.imshow(
    correlation,
    text_auto=".2f",
    color_continuous_scale="RdBu_r",
    zmin=-1,
    zmax=1,
    title="Bengaluru House Data - Correlation Heatmap",
    labels={
        "x": "Features",
        "y": "Features",
        "color": "Correlation"
    }
)
# 6. Customize chart
fig.update_layout(
    width=900,
    height=700,
    title_x=0.5
)
# 7. Display interactive chart
fig.show()
# 8. Export to HTML
fig.write_html("bengaluru_interactive_heatmap.html")
print("Interactive heatmap created successfully!")
```

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ANNEXURE C: PYTHON SOURCE CODE — BBMP CHOROPLETH MAPPING

C.1 Ward-Name Normalization, Spatial Join & Plotly Choropleth (`project3/BBMP chloropleth/chloropleth.py`)

```
def normalize(value):
    if pd.isna(value):
        return ""
    value = unicodedata.normalize("NFKD", str(value))
    value = value.lower().strip()
    value = re.sub(r"\bward\b", "", value)
    value = re.sub(r"[^a-z0-9]", "", value)
    return value
# Find possible ward-related columns
candidates = [
    c for c in df.columns
    if any(word in c.lower() for word in
           ["ward", "name", "number", "no."])
]
properties = list(geojson["features"][0]["properties"].keys())
if not candidates:
    raise ValueError(
        "Could not identify a ward column in the CSV. "
        "Check the printed CSV columns."
    )
# Find the best matching pair of ward fields
best_match = None
for prop in properties:
    geo_names = {
        normalize(feature["properties"].get(prop))
        for feature in geojson["features"]
    }
    geo_names.discard("")
    for col in candidates:
        csv_names = df[col].map(normalize)
        overlap = csv_names[csv_names.isin(geo_names)].nunique()
        if best_match is None or overlap > best_match[0]:
            best_match = (overlap, prop, col)
if best_match is None or best_match[0] == 0:
    raise ValueError(
        "No ward names matched. Compare the CSV columns and "
        "GeoJSON properties printed above."
    )
match_count, geo_ward_field, csv_ward_field = best_match
print("\nGeoJSON ward field:", geo_ward_field)

# ... (Metric selection & duplicate ward validation) ...

fig = px.choropleth_map(
    map_df,
    geojson=matched_geojson,
    locations="_ward_key",
    featureidkey="properties._ward_key",
    color=metric,
    hover_name=csv_ward_field,
    hover_data={
        metric: True,
        "_ward_key": False
    },
    color_continuous_scale="YlGnBu",
    map_style="carto-positron",
    center={"lat": 12.9716, "lon": 77.5946},
    zoom=9,
    opacity=0.78,
    title=f"BBMP Ward-wise Distribution of {metric}"
)
fig.update_layout(
    title_x=0.5,
    margin={"r": 10, "t": 60, "l": 10, "b": 10},
    coloraxis_colorbar={
        "title": metric
    }
)
# Display map
fig.show()
# ------------------------------------------------------------
# 8. EXPORT THE INTERACTIVE MAP TO HTML
# ------------------------------------------------------------
output_file = data_dir / "BBMP_Public_Goods_Choropleth.html"
fig.write_html(
    output_file,
    include_plotlyjs=True,
    full_html=True
```

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ANNEXURE D: CAPSTONE CHART ENGINE & POWER BI DAX CODE

D.1 Encroachment vs. Biodiversity Regression (`backend/chart_generator.py`)

```
def generate_chart_encroachment_correlation(df=None, save_path=None):
    if df is None: df = load_data()
    fig, ax = plt.subplots(figsize=(9.5, 5.2), facecolor='#FFFFFF')
    sns.regplot(
        data=df,
        x='encroachment_pct',
        y='biodiversity_index',
        scatter_kws={'alpha': 0.65, 'color': '#2874A6', 's': 35},
        line_kws={'color': '#C0392B', 'linewidth': 2},
        ax=ax
    )
    # Calculate Pearson r
    r = df['encroachment_pct'].corr(df['biodiversity_index'])
    ax.set_title(f"Impact of Lake Bed Encroachment on Avian & Floral Biodiversity (r = {r:.
2f})\nHigh perimeter encroachment correlates directly with ecological barrenness",
                 fontsize=11.5, fontweight='bold', pad=14, loc='left', color='#1B4F72')
    ax.set_xlabel("Perimeter Encroachment Rate (%)", fontsize=10, labelpad=8)
    ax.set_ylabel("Biodiversity Index (0–100 Scale)", fontsize=10, labelpad=8)
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)
    # Annotation on the regression
    ax.annotate("Fenced & Protected Lakes\n(Jakkur, Agara, Kaikondrahalli)\nHigh Bird Nesting (
>85)",
                xy=(1.0, 90.0), xytext=(12.0, 88.0),
                arrowprops=dict(facecolor='#1B4F72', arrowstyle='->', lw=1.2),
                fontsize=8.5, fontweight='bold', color='#1B4F72',
                bbox=dict(boxstyle="round,pad=0.3", fc="#EBF5FB", ec="#AED6F1", lw=0.8))
    ax.annotate("Severe Encroachment Zone\nUnchecked commercial & construction debris\
nBiodiversity collapses below 30",
                xy=(45.0, 22.0), xytext=(35.0, 45.0),
                arrowprops=dict(facecolor='#C0392B', arrowstyle='->', lw=1.2),
                fontsize=8.5, fontweight='bold', color='#C0392B',
                bbox=dict(boxstyle="round,pad=0.3", fc="#FDEDEC", ec="#F5B7B1", lw=0.8))
    plt.tight_layout()
    if save_path:
        plt.savefig(save_path, dpi=200, bbox_inches='tight')
        plt.close()
        return save_path
```

D.2 Pithambar's Namma Metro DAX Measures (`project2/DAX measures/README.md`)

```
-- Pithambar's Assigned DAX Measures (Project 2: Namma Metro Power BI)
Average Ridership = AVERAGE(Fact_Ridership[Ridership])

Peak Hour Ridership =
MAXX(
    ALL(Dim_Hour[Value]),
    CALCULATE([Total Ridership])
)

Peak Hour Share =
DIVIDE(
    [Peak Hour Ridership],
    [Total Ridership],
    0
)
```